9/10/26

genetic optimization Algorithm

pseudo code

1. initialize a proposition of random chromosome
2. evaluate the fitness of each chromosome
3. while termination condition is not satisfied

do

a. Select parents based on fitness

b. perform crossover on selected parents

c. perform mutation on offspring

d. evaluate the fitness of the offspring

e. replace some (or) all individuals in the population with offspring

END WHILE

4 Return the best solution (chromosome)

END

Agricultural Irrigation Scheduling

Determine Irrigation Schedule based on crop water requirement and resource available

1) minimize water usage while maintaining crops

Pseudocode

Algorithm (Generic_Algorithm_Irrigation)

Define crop water requirement

Define available amount of water and time slots.

Algorithm

(Genetic_Algorithm_Irrigation)

Begin

Input Crop water requirements

Input available amount of water

Input the irrigation time slots

Set population size

Set crossover rate

Set mutation rate

Generate random irrigation schedule

for each irrigation do

Calculate total water used

Calculate water storage

Check resource constraints

Calculate fitness

end for

from generation = 1 to maximum Do

Select parent schedule based on fitness

Apply crossover to produce offspring

Apply mutation to offspring

for each offspring check

water availability

check crop water requirements

calculate total water usage

Calculate fitness

end for

From next generation update best feasible

Schedule

return the output best irrigation scheduling

Output

Best irrigation Schedule :

Crop A: 21 litres

Crop B: 30 litres

Crop C: 26 litres

total water used : 77 litres

water available : 80 litres

Fitness : 77

O/P Seen
Sneha
09/10/26

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